StockSense AI

XGBOOST-BASED FUNDAMENTAL STOCK SCREENING AND LONG-TERM INVESTMENT RECOMMENDATION SYSTEM

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Problem Statement

Long-term investors usually check whether a company is fundamentally strong before buying a stock. This requires reading many financial ratios such as ROE, ROCE, debt-to-equity, P/E, growth, free cash flow, promoter holding, and pledge percentage. For beginners and family investors, this process is slow, confusing, and dependent on personal experience.

Project goal

- collect important company fundamentals,
- analyze long-term fundamental strength,
- generate a 0–100 AI strength score,
- classify stocks as Buy, Watchlist, or Avoid,
- provide beginner-friendly guidance and risk warnings.

What Was Implemented from Research Papers

- From fundamental-analysis ML papers: use financial indicators instead of only price charts.
- From stock recommendation papers: convert financial indicators into ranking/recommendation output.
- From finance-ML review papers: consider model evaluation, data quality, and interpretability.
- From practical decision-support systems: provide a user-facing workflow, not only a model result.

What is added beyond the papers

- beginner guide videos for collecting values,
- 15-parameter web-based input form,
- XGBoost-powered score generation,
- explainable category-level diagnostics,
- macro-risk alerts for wars, tariffs, rate hikes, and policy shocks.

Research Paper Comparison

Paper	Core idea	Data type	Relevance to project
Huang et al. (IEEE SSCI 2021)	ML for stock prediction based on fundamental analysis	Quarterly financial data	Supports use of fundamentals for ML-assisted investment decisions
Yang et al. (arXiv 2025)	Dynamic stock recommendation using financial indicators and model selection	Stock indicators and returns	Matches recommendation/ranking direction
Saberironaghi et al. (AppliedMath 2025)	Review of ML and deep learning in stock market prediction	Survey paper	Helps justify ML use and discuss limitations

Project positioning: StockSense AI is closer to fundamental stock screening and recommendation than short-term stock price prediction.

Dataset Used

The project currently uses a synthetic expert-rule dataset designed to imitate multiple long-term stock fundamental profiles. Each row represents one company-year style snapshot with 15 financial indicators and one target score.

Processed dataset summary

- total rows: **62,500**
- feature columns: **15**
- target column: **score** from 0 to 100
- output classes: **Buy, Watchlist, Avoid**

Score distribution

- Avoid (< 60): **24,937**
- Watchlist ($60--79$): **18,820**
- Buy (≥ 80): **18,743**

Synthetic Dataset Coverage

The dataset was strengthened to handle many real-world-like cases, not only ideal examples.

- quality compounders,
- average companies,
- expensive quality stocks,
- cyclicals,
- distressed companies,
- debt traps,
- pledge-risk companies,
- turnaround cases,
- negative cash-flow growth cases,
- zero-debt high coverage cases.

Important note: The dataset is strong for hackathon demonstration and repeatable training, but real-world investment-performance claims require historical fundamentals and future return validation.

Input Parameters

Valuation

- Stock P/E
- Industry P/E
- Price to book value

Profitability

- ROE
- ROCE
- OPM

Safety

- Debt to equity
- Interest coverage
- Current ratio

Growth

- Sales growth 5 years
- Profit growth 5 years
- EPS growth 3 years

Cash and ownership

- Free cash flow
- Promoter holding
- Promoter pledge

End-to-End Architecture

Layer	Function
Frontend	Guide videos, 15-input analysis form, progress indicator, result gauge, recommendation page
Flask backend	Validates inputs, loads feature order, prepares model input, handles errors, exposes health route
ML engine	XGBoost regressor predicts a 0–100 fundamental strength score
Explanation layer	Groups signals into valuation, profitability, balance-sheet safety, growth, and ownership/cash quality
Risk guidance	Warns users to re-evaluate during wars, tariffs, interest-rate hikes, economic shocks, or major policy changes

User fundamentals → Flask validation → XGBoost model → Score → Buy / Watchlist / Avoid

Preprocessing, Training and Evaluation Pipeline

Dataset generation

1. define fundamental indicators and expert-rule relations,
2. generate multiple company regimes and edge cases,
3. assign a 0–100 score using weighted rules and red-flag penalties,
4. store final dataset in `data/training_data.csv`.

Training

1. load 15 feature columns,
2. clean missing and infinite values,
3. split into train and test data,
4. train XGBoost regressor,
5. save `stocksense_model.pkl`.

Evaluation

1. compute MAE and R^2,
2. convert score into Buy, Watchlist, Avoid buckets,
3. compute bucket accuracy,
4. export model metadata and feature importance.

Scoring Logic

The target score is generated using a weighted fundamental-analysis philosophy. Positive signals increase the score, while red flags reduce the score.

Positive signals

- high ROE and ROCE,
- consistent sales, profit, and EPS growth,
- low debt-to-equity,
- strong interest coverage,
- positive free cash flow,
- high promoter holding and low promoter pledge.

Red flags

- excessive valuation compared with industry P/E,
- high leverage,
- weak interest coverage,
- negative free cash flow,
- high promoter pledge,
- poor liquidity or weak profitability.

XGBoost Model

Model used: `XGBRegressor`

Why XGBoost was selected

- works well on structured tabular financial data,
- captures non-linear relationships between fundamentals,
- handles interactions such as high growth with high valuation,
- gives feature importance for explainability,
- fast enough for web application inference.

Training configuration

- estimators: 650
- max depth: 4
- learning rate: 0.04
- subsample: 0.85
- column sample by tree: 0.85
- regularization: alpha 0.05, lambda 1.5

How Recommendation is Generated

The model predicts a continuous score from 0 to 100. The web application then converts the score into an investor-friendly recommendation.

Score range	Output	Meaning
≥ 80	Buy	Strong long-term candidate according to entered fundamentals
60–79	Watchlist	Moderate fundamentals; monitor and re-analyze later
< 60	Avoid	Weak or risky fundamentals; avoid unless fundamentals improve

Current XGBoost Analytics

From `model_metadata.json`

- Rows after cleaning: **62,500**
- Model: **XGBRegressor**
- Mean Absolute Error: **2.80**
- R^2 score: **0.982**
- Buy/Watchlist/Avoid bucket accuracy: **91.45%**

Interpretation

- MAE of 2.80 means the predicted score is usually within about 3 points of the synthetic label.
- $R^2 = 0.982$ indicates strong fit to the generated fundamental scoring logic.
- Bucket accuracy measures whether the recommendation class is correct.

Feature Importance

Feature	Importance	Interpretation
ROCE	0.302	Most important capital-efficiency signal
ROE	0.211	Strong shareholder-return signal
Profit growth 5Y	0.115	Measures long-term business growth
Interest coverage	0.076	Captures debt servicing safety
Sales growth 5Y	0.073	Revenue consistency signal
Current ratio	0.051	Liquidity and short-term safety
Free cash flow	0.044	Cash-generation quality

Interpretation: The model focuses mainly on profitability, capital efficiency, growth, and balance-sheet safety, which matches long-term fundamental analysis.

Website Flow

1. User opens StockSense AI homepage.
2. User watches guide videos to understand how to collect values.
3. User enters all 15 fundamental parameters.
4. Flask validates every value.
5. XGBoost model generates a score.
6. Result page shows:
   - score gauge,
   - Buy / Watchlist / Avoid recommendation,
   - strength signals,
   - review-before-buying signals,
   - risk alerts and news-monitoring guidance.

How This Project Improves on Basic ML Demos

- Many stock ML projects focus only on price prediction.
- This project focuses on fundamental long-term screening.
- It combines:
  - financial-ratio based input,
  - XGBoost model prediction,
  - beginner-friendly frontend,
  - explanation layer,
  - model metadata and feature importance,
  - risk warnings and re-analysis guidance.
- So the project is an end-to-end decision-support application, not only a model notebook.

Important Limitation

Do not overclaim

The current model is trained on a synthetic expert-rule dataset. Therefore, the accuracy shows how well the model learned the generated fundamental scoring logic. It does not prove real-world stock return prediction accuracy.

Correct claim

- The XGBoost model achieved **91.45%** Buy/Watchlist/Avoid bucket accuracy on a 62,500-row synthetic fundamental-analysis dataset.

Incorrect claim

- The model predicts real stock market returns with 91.45% accuracy.

Future Improvements

- collect real historical fundamentals for NSE-listed companies,
- add 1-year, 3-year, and 5-year future return labels,
- validate the model on unseen real companies,
- add sector-wise normalization,
- add peer comparison,
- add SHAP-based model explanations,
- integrate compliant finance APIs where available,
- improve UI with portfolio watchlist and saved analysis history.

Conclusion

StockSense AI is a fundamental-analysis based stock screening system for long-term investors. It converts 15 important financial indicators into a 0–100 strength score using an XGBoost model and presents the output as Buy, Watchlist, or Avoid.

The project includes a beginner-friendly Flask web application, guide videos, input validation, explainable category diagnostics, feature-importance reporting, and macro-risk warnings. The current XGBoost model performs strongly on a 62,500-row synthetic expert-rule dataset.

The next major improvement is to validate the same system on real historical fundamentals and future stock returns, which would make the project stronger for research and real-world investment decision support.

References

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Thank You